

Resume Analyzer with Job Matching

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Introduction

In this project, we created a smart system that analyzes resumes and matches them with the most suitable job roles. Using Python, we built a tool that reads resumes in DOCX format, extracts important skills, and uses a Decision Tree model to predict the best job for each candidate based on their skills.

Approach

The system follows a modular and object-oriented structure divided into four main components:

1. **ResumeParser**: Extracts candidate information such as name, job title, and skills from `.docx` files.
2. **ModelTrainer**: Trains a `DecisionTreeClassifier` on parsed data, transforming skill sets into binary format.
3. **ResumeMatcher**: Loads the model and predicts job titles based on skill matching. Calculates match percentages and suggests alternatives.
4. **Visualization and Evaluation**: Displays performance using pie and bar charts, and visualizes the decision tree.
5. **ResumeTester (Optional)**: Accepts manual user input and predicts job titles in real-time.

Challenges

Several challenges were encountered:

- Parsing resumes with inconsistent formats.
- Maintaining model accuracy with a limited dataset.
- Designing meaningful skill matching logic.
- Handling errors such as unreadable files or missing data gracefully.

Conclusion

This Resume Analyzer project showcases the power of automation and machine learning in the hiring process. The system efficiently processes resumes, predicts job roles, and visualizes results. Despite data limitations, it provides a structured and extendable solution for smarter recruitment workflows.

Team Members

Output

Model trained and saved successfully.

Resume Matching Results:

Name	Actual Job	Predicted Job	Match %	Correct Match
Youssef Al-Rashid	Human Resources Specialist	Human Resources Specialist	83.33%	✓
Omar Al-Qadhi	Software Engineer	Software Engineer	83.33%	✓
Noura Al-Fahad	Graphic Designer	Graphic Designer	0%	✗
Ahmed Al-Rashid	Accountant	Accountant	50.0%	✗
Aisha Al-Harbi	Accountant	Accountant	100.0%	✓
Fatimah Al-Fahad	Software Engineer	Software Engineer	100.0%	✓
Mariam Al-Johani	Human Resources Specialist	Human Resources Specialist	83.33%	✓
Youssef Al-Johani	Software Engineer	Software Engineer	83.33%	✓
Omar Al-Shammari	Human Resources Specialist	Human Resources Specialist	66.67%	✓
Layla Al-Fahad	Accountant	Accountant	66.67%	✓
Mariam Al-Mansoori	Human Resources Specialist	Human Resources Specialist	83.33%	✓
Mohammed Al-Qadhi	Accountant	Accountant	100.0%	✓
Omar Al-Mansoori	Marketing Manager	Marketing Manager	66.67%	✓
Ali Al-Harbi	Marketing Manager	Marketing Manager	83.33%	✓
Fatimah Al-Harbi	Graphic Designer	Graphic Designer	83.33%	✓
Ali Al-Saadi	Accountant	Accountant	50.0%	✗
Fatimah Al-Qadhi	Software Engineer	Software Engineer	100.0%	✓
Noor Al-Shammari	Systems Analyst	Systems Analyst	100.0%	✓
Noor Al-Rashid	Systems Analyst	Systems Analyst	66.67%	✓
Ali Al-Mansoori	Human Resources Specialist	Human Resources Specialist	100.0%	✓
Youssef Al-Mansoori	Accountant	Accountant	100.0%	✓
Mohammed Al-Dosari	Accountant	Accountant	66.67%	✓

Figure 1: Resume Matching Results (Page 1)

Fatimah Al-Shammari	Marketing Manager	Marketing Manager	83.33%	✓
Omar Al-Harbi	Graphic Designer	Graphic Designer	50.0%	✗
Youssef Al-Harbi	Accountant	Accountant	66.67%	✓
Fatimah Al-Shammari	Software Engineer	Software Engineer	83.33%	✓
Layla Al-Mansoori	Marketing Manager	Marketing Manager	83.33%	✓
Youssef Al-Dosari	Accountant	Accountant	83.33%	✓
Ahmed Al-Dosari	Human Resources Specialist	Human Resources Specialist	83.33%	✓
Mohammed Al-Ghamdi	Software Engineer	Software Engineer	66.67%	✓
Youssef Al-Ghamdi	Human Resources Specialist	Human Resources Specialist	100.0%	✓
Ali Al-Shammari	Graphic Designer	Graphic Designer	50.0%	✗
Noor Al-Shammari	Marketing Manager	Marketing Manager	83.33%	✓
Omar Al-Fahad	Marketing Manager	Marketing Manager	66.67%	✓
Omar Al-Fahad	Graphic Designer	Graphic Designer	83.33%	✓
Mohammed Al-Mansoori	Accountant	Accountant	66.67%	✓
Ahmed Al-Shammari	Marketing Manager	Marketing Manager	83.33%	✓
Noor Al-Fahad	Software Engineer	Software Engineer	66.67%	✓
Ahmed Al-Qadhi	Marketing Manager	Marketing Manager	66.67%	✓
Reem Al-Harbi	Systems Analyst	Graphic Designer	0%	✗
Mariam Al-Rashid	Software Engineer	Software Engineer	83.33%	✓
Huda Al-Harbi	Marketing Manager	Graphic Designer	0%	✗
Aisha Al-Saadi	Software Engineer	Software Engineer	83.33%	✓
Layla Al-Fahad	Accountant	Accountant	83.33%	✓
Ahmed Al-Dosari	Accountant	Accountant	83.33%	✓
Mariam Al-Dosari	Systems Analyst	Systems Analyst	66.67%	✓
Ali Al-Fahad	Accountant	Accountant	66.67%	✓
Ali Al-Johani	Graphic Designer	Graphic Designer	50.0%	✗
Noor Al-Shammari	Systems Analyst	Systems Analyst	66.67%	✓
Youssef Al-Johani	Systems Analyst	Systems Analyst	100.0%	✓

Figure 2: Resume Matching Results (Page 2)

Ali Al-Saadi	Human Resources Specialist	Human Resources Specialist	50.0%	X
Omar Al-Qadhi	Marketing Manager	Marketing Manager	66.67%	✓
Noor Al-Ghamdi	Accountant	Accountant	50.0%	X
Ahmed Al-Dosari	Graphic Designer	Graphic Designer	83.33%	✓
Omar Al-Fahad	Graphic Designer	Graphic Designer	83.33%	✓
Ahmed Al-Harbi	Graphic Designer	Graphic Designer	83.33%	✓
Ali Al-Harbi	Graphic Designer	Graphic Designer	83.33%	✓
Mariam Al-Saadi	Accountant	Accountant	83.33%	✓
Layla Al-Fahad	Accountant	Accountant	66.67%	✓
Ali Al-Mansoori	Systems Analyst	Systems Analyst	100.0%	✓
Ahmed Al-Rashid	Systems Analyst	Systems Analyst	100.0%	✓
Huda Al-Rashid	Software Engineer	Graphic Designer	0%	X
Fatimah Al-Qadhi	Systems Analyst	Systems Analyst	66.67%	✓
Layla Al-Mansoori	Systems Analyst	Systems Analyst	100.0%	✓
Noor Al-Qadhi	Graphic Designer	Graphic Designer	66.67%	✓
Layla Al-Shammari	Marketing Manager	Marketing Manager	83.33%	✓
Noor Al-Qadhi	Systems Analyst	Systems Analyst	66.67%	✓
Mariam Al-Saadi	Accountant	Accountant	83.33%	✓
Layla Al-Mansoori	Human Resources Specialist	Human Resources Specialist	50.0%	X
Youssef Al-Harbi	Systems Analyst	Systems Analyst	83.33%	✓
Noor Al-Johani	Software Engineer	Software Engineer	66.67%	✓
Noor Al-Saadi	Accountant	Accountant	66.67%	✓
Layla Al-Qadhi	Software Engineer	Software Engineer	100.0%	✓
Mariam Al-Rashid	Systems Analyst	Systems Analyst	50.0%	X
Ahmed Al-Qadhi	Human Resources Specialist	Human Resources Specialist	83.33%	✓
Ali Al-Johani	Graphic Designer	Graphic Designer	50.0%	X
Mohammed Al-Rashid	Marketing Manager	Marketing Manager	66.67%	X
Mohammed Al-Johani	Human Resources Specialist	Human Resources Specialist	50.0%	X

Figure 3: Resume Matching Results (Page 3)

Mohammed Al-Johani	Marketing Manager	Marketing Manager	50.0%	X
Noor Al-Mansoori	Software Engineer	Software Engineer	100.0%	✓
Aisha Al-Shammari	Human Resources Specialist	Human Resources Specialist	100.0%	✓
Layla Al-Harbi	Human Resources Specialist	Human Resources Specialist	66.67%	✓
Noor Al-Qadhi	Accountant	Accountant	100.0%	✓
Youssef Al-Fahad	Accountant	Accountant	66.67%	✓
Noor Al-Johani	Accountant	Accountant	66.67%	✓
Aisha Al-Saadi	Systems Analyst	Systems Analyst	83.33%	✓
Layla Al-Mansoori	Accountant	Accountant	100.0%	✓
Aisha Al-Harbi	Graphic Designer	Graphic Designer	0%	X
Fatimah Al-Dosari	Graphic Designer	Graphic Designer	66.67%	✓
Layla Al-Ghamdi	Graphic Designer	Graphic Designer	66.67%	✓
Mariam Al-Johani	Accountant	Accountant	83.33%	✓
Youssef Al-Dosari	Marketing Manager	Marketing Manager	66.67%	X
Omar Al-Shammari	Accountant	Accountant	100.0%	✓
Mariam Al-Rashid	Systems Analyst	Systems Analyst	66.67%	✓
Mariam Al-Harbi	Human Resources Specialist	Human Resources Specialist	83.33%	✓
Salem Al-Harbi	Systems Analyst	Graphic Designer	0%	X
Youssef Al-Fahad	Accountant	Accountant	100.0%	✓
Omar Al-Fahad	Marketing Manager	Graphic Designer	0%	X
Omar Al-Qadhi	Systems Analyst	Systems Analyst	100.0%	✓
Omar Al-Ghamdi	Human Resources Specialist	Human Resources Specialist	66.67%	✓
Layla Al-Johani	Accountant	Accountant	100.0%	✓

Figure 4: Resume Matching Results (Page 4)

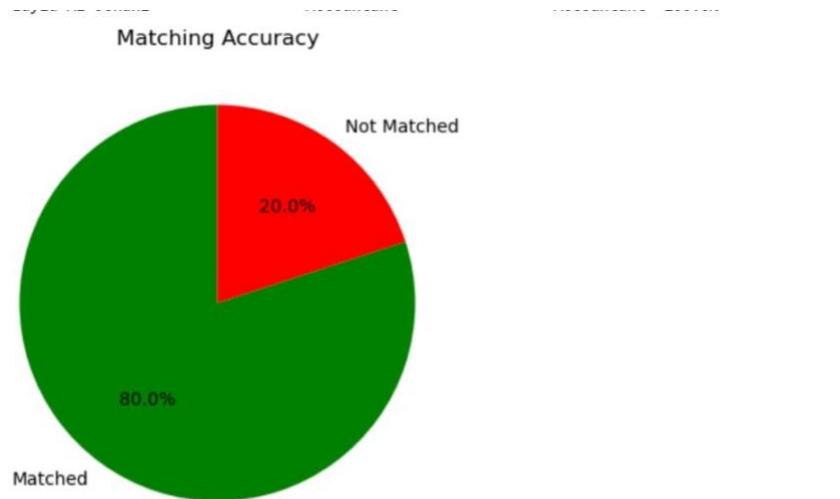


Figure 5: Matching Accuracy Pie Chart

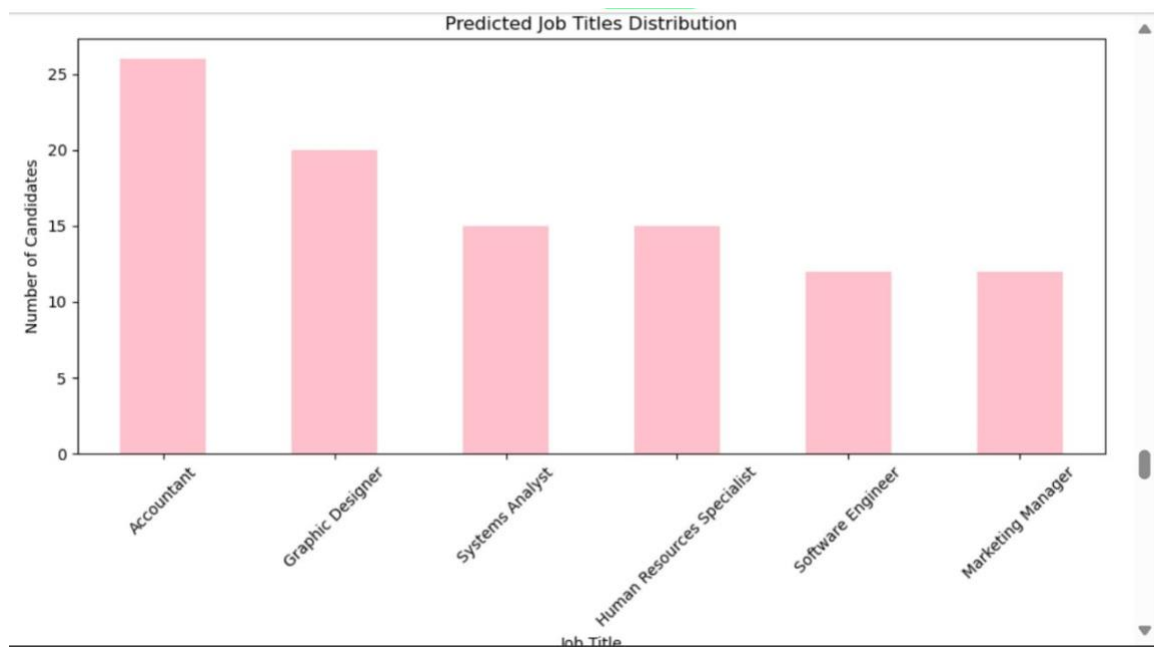


Figure 6: Predicted Job Titles Distribution

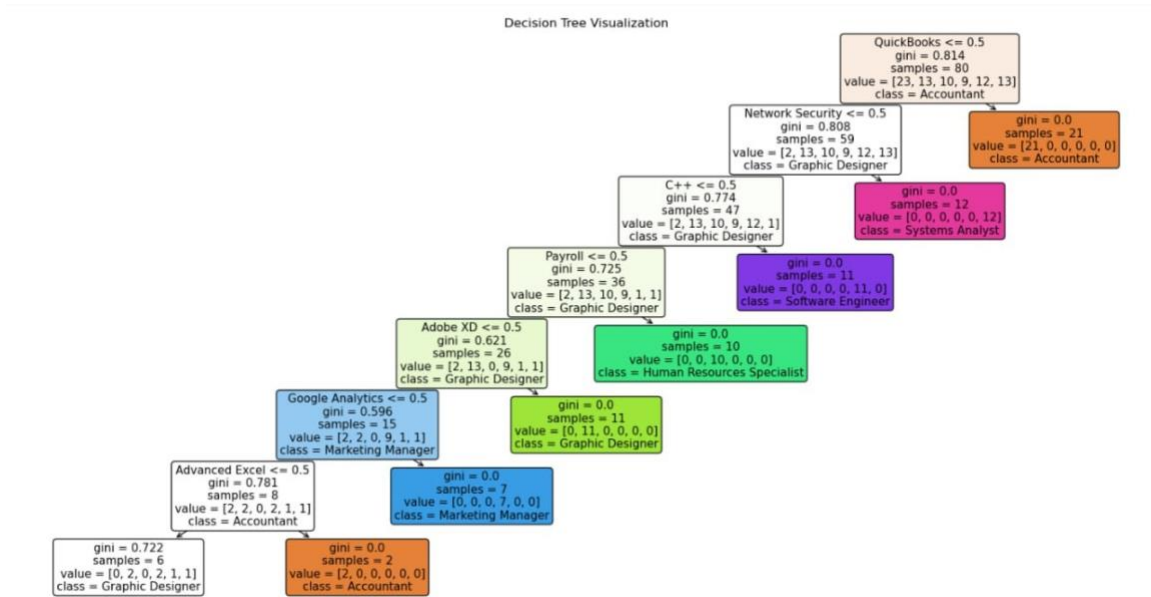


Figure 7: Decision Tree Visualization

Enter your name: ruba
 Enter your email: ihhiuyu@gmail.com
 Enter your skills (comma-separated): Payroll,Labor Law

ruba , based on your skills, the suggested job is: Human Resources Specialist

Figure 8: Manual User Input Result

Team Members

Name	Student ID
Dayala Baslaam	2410447
Ruba Alabsi	2410691

Code Task Distribution

Task	Dayala Baslaam	Ruba Alabsi
Resume Parsing (ResumeParser)	✓	
Model Training (ModelTrainer)	✓	
Prediction Logic (ResumeMatcher)		✓
Decision Tree Implementation		✓
Visualization & Plotting	✓	
Error Handling and Testing		✓